**CPSC 459 Project 2**

**Extra credit**

You are to use Simulation.java to analyze p_malicious along with other parameters.

Use the following commands to compile and run **Simulation** in the grading folder.

$ javac Simulation.java

$ java Simulation .1 .15 .01 10

**Change values for the four required command line arguments** for all 3x3x3x2 = 54 combinations

parameter for random graph: prob. that an edge will exist
p_graph (.1, .2, .3)

prob. that a node will be set to be malicious
p_malicious (.15, .30, .45)

probability of assigning an initial transaction to each node
p_txDistribution (.01, .05, .10)

number of simulation rounds your nodes will run for
numRounds (10, 20)

You will get the result for a set of transactions that each node believes consensus on every time you run Simulation.

**Note that: All the compliant nodes need to believe the consensus, otherwise the percentage of malicious nodes have NOT been tolerated!**

**Write a report** based on the results to show the maximum percentage of malicious nodes that can be tolerated given the other three parameters.

Your report should contain information that looks similar to the following:

| the maximum percentage of malicious nodes p_malicious | prob. that an edge will exist p_graph | probability of assigning an initial transaction to each node p_txDistribution | number of simulation rounds your nodes will run for numRounds |
|---|---|---|---|
| 30% | 30% | 1% | 20 |
| … | … | … | … |

**Deliverable**:
Submit a ***extra credit.pdf*** file for extra credit.